

Question	Answer	Marks
1(a)	arrow vertically downwards labelled 'weight' <u>and</u> arrow perpendicular to cone, inwards and upwards, labelled 'normal contact force'	B1
1(b)	vertical component of contact force = weight (so no resultant force vertically)	B1
	horizontal component of contact force is resultant force towards centre (of circle)	B1
1(c)	$a = v^2 / r$	C1
	$a = g \tan 52^\circ$	C1
	$9.81 \times \tan 52^\circ = v^2 / 0.15$ leading to $v = 1.4 \text{ m s}^{-1}$	A1
1(d)	$v = r\omega$ <u>or</u> $a = r\omega^2$	C1
	$\omega = 1.4 / 0.15$ or $\sqrt{(9.81 \times \tan 52^\circ / 0.15)}$ $= 9.3 \text{ rad s}^{-1}$	A1
1(e)	same resultant force / same acceleration so v^2 is proportional to r (so if speed increases radius must also increase)	A1

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